

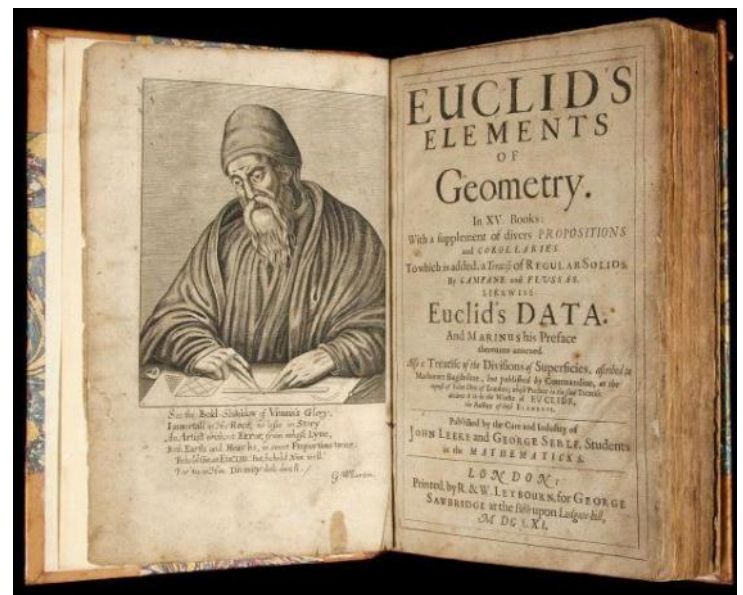
Geometry

Friday, May 17, 2024

11:44 AM

Euclid

- The Father of Geometry
- Lived 325 - 265 BC in Alexandria, Egypt
- Wrote "The Elements," his treatise of Geometry
- He defined many "axioms," i.e., statements considered true but are not proven



Axioms

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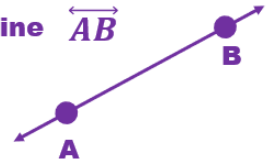
Some Axioms

- Point - that which has no part, no length, width, depth, but just a location in space. In Math, points are defined by a Capital letter (A, B, C, ...) or a Capital Letter with a numeric subscript (P_1 , P_2 , P_3 , etc.).
- Line - a straight path of points that continue infinitely in opposite directions. In Math, lines are defined by any two points through which the line passes with a double arrow line above the points ($\overleftrightarrow{AB}$).
- Ray - a straight path of points with one terminal point and the other end that continues infinitely. A ray is also defined by two points, the terminal point and any other point, and a single arrow above the points ($\overrightarrow{AB}$).
- Line Segment - a straight path of points with two terminal points. A line segment is defined by its terminal points and a line above the points ($\overline{AB}$).

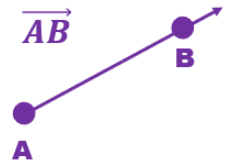
Point A



Line $\overleftrightarrow{AB}$



Ray $\overrightarrow{AB}$



Line Segment $\overline{AB}$



Lines

Saturday, May 18, 2024 7:49 AM

**Horizontal
Line**



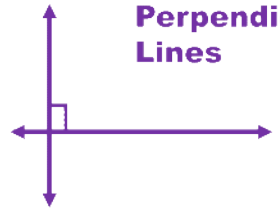
**Vertical
Line**



**Parallel
Lines**

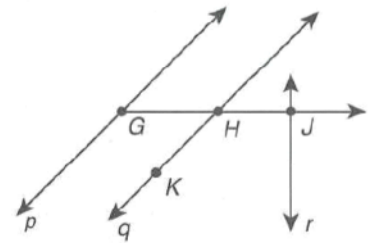


**Perpendicular
Lines**



Answer the questions about the drawing to the right.

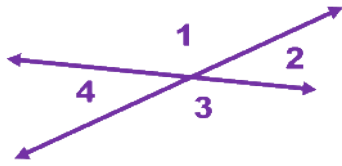
1. How many lines are shown in the drawing?
2. Point J lies at the intersection of perpendicular lines. Are lines r and p perpendicular? Explain your reasoning.
3. Name 4 line segments found in the drawing.
4. There are many rays in the drawing. Name two.



Angles

Friday, May 17, 2024 11:48 AM

Lines that intersect cross at exactly 1 point. The intersection of two lines creates 4 angles.



Vertical Angles are opposite each other $\angle 4$ & $\angle 2$ or $\angle 1$ & $\angle 3$

Acute angle $\angle A < 90^\circ$

Right angle $\angle A = 90^\circ$

Obtuse angle $\angle A > 90^\circ$

Straight angle $\angle A = 180^\circ$ this is a straight line

Complementary angles $\angle A + \angle B = 90^\circ$

Supplementary angles $\angle A + \angle B = 180^\circ$

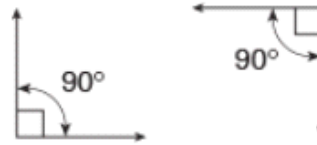
Angles, cont'd...

Monday, May 27, 2024 6:53 PM

Name of Angle Properties

Examples

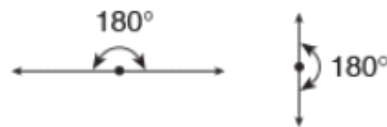
right angle exactly 90°



acute angle less than 90°



straight angle exactly 180°



obtuse angle between 90° and 180°

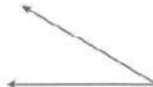


Answer each question.

5. Match the illustrations with the descriptions.

- _____ an acute angle
- _____ a right angle
- _____ an obtuse angle
- _____ a straight angle

a.  b. 

c.  d. 

6. Which of the three angles is smallest? Why?
(*Hint:* The size of an angle is based only on the angle, not the length of the sides.)

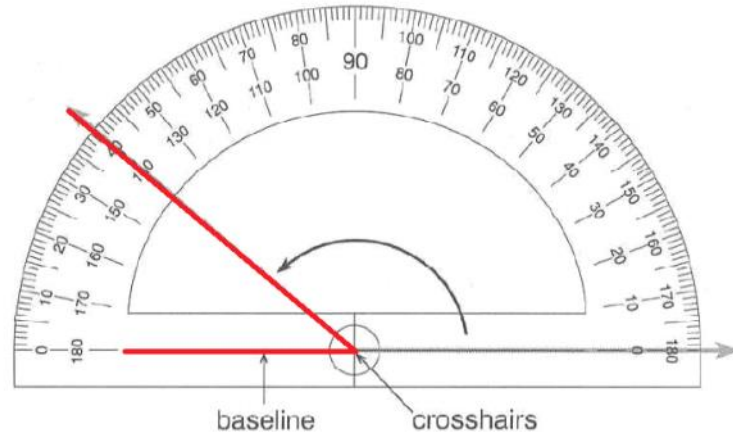
a.  b.  c. 

USING PROTRACTORS

A **protractor** is a tool for measuring angles. A protractor looks like a half circle with a fanlike scale of numbers. One scale runs from left to right; the other from right to left.

To measure an angle, line up one ray of the angle with the baseline of the protractor. Put the center point of the protractor (sometimes marked with crosshairs) on the vertex of the angle. Read the scale at the point where the other ray crosses it.

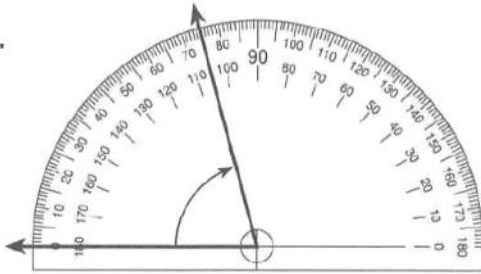
This illustration shows a protractor and an angle of 140° . The symbol $\angle$ means angle. Angles can be named by the vertex point, by a letter or number within the opening, or by three points.



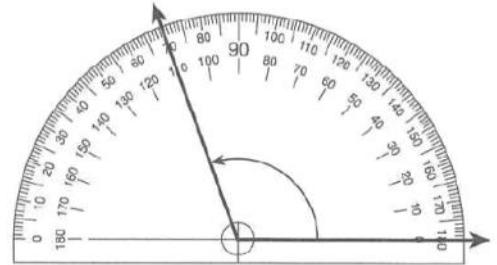
Answer each question.

1. Tell the number of degrees in each of the following angles.

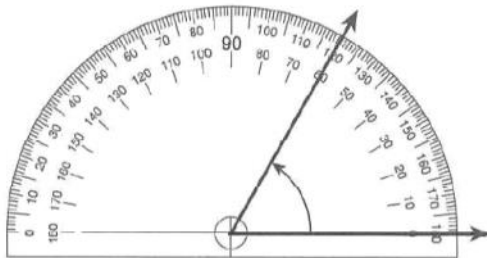
a.



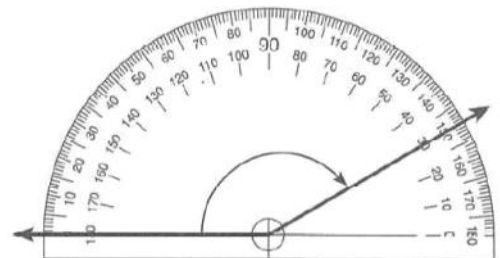
c.



b.



d.



2. Match the angle with the correct number of degrees. Use your judgment or a protractor if you have one.

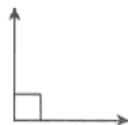
_____ 180°

_____ 90°

_____ 30°

_____ 135°

a.



b.



c.



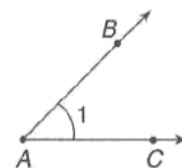
d.



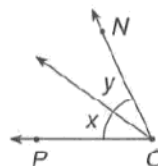
Naming Angels & Arithmetic

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The angle at the right could be named $\angle A$, $\angle 1$, $\angle BAC$, or $\angle CAB$. When three points are used to name an angle, the vertex is always the middle letter.



In this illustration, $\angle NOP$ is divided into two smaller angles, $\angle x$ and $\angle y$. If $\angle x = 35^\circ$ and $\angle y = 30^\circ$, we can add to find the measure of $\angle NOP$.



$$35^\circ + 30^\circ = 65^\circ$$

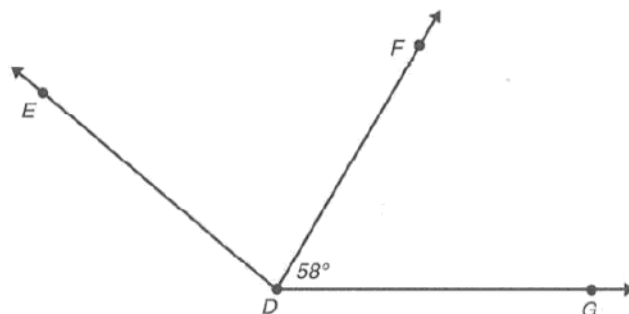
$$\angle NOP = 65^\circ$$

Subtraction can be used to find a missing angle.
(**Note:** The m before the angle symbol $\angle$ means "the measure of.")

$$m\angle EDG = 140^\circ. \text{ Find } m\angle EDF.$$

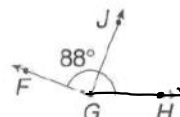
$$m\angle EDG - m\angle FDG = m\angle EDF$$

$$140^\circ - 58^\circ = 82^\circ$$



Solve the following problems without using a protractor.

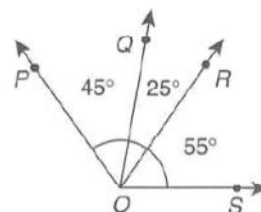
3. $\angle FGH = 160^\circ$. Find the measure of $\angle JGH$.



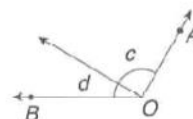
4. Tell the type of angle and number of degrees for each angle:

Angle	Type	Degrees
$\angle POS$		
$\angle QOS$		
$\angle POR$		

Hint: never trust a picture - determine degrees before identifying the type of angle.



5. $\angle AOB = 120^\circ$ and $\angle c$ is 3 times as great as $\angle d$. Use algebra to find the measurements of both $\angle c$ and $\angle d$. (**Hint:** Let a variable equal the measurement of $\angle d$.)

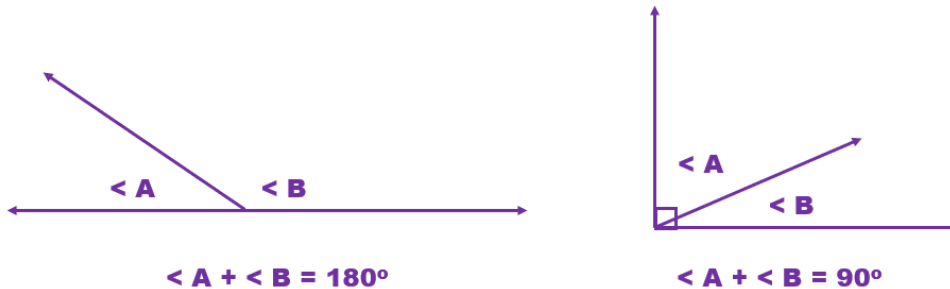


Special Angles

Wednesday, May 29, 2024 7:40 PM

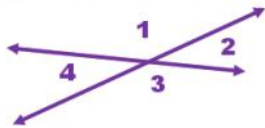
Recall:

- If $\angle A$ and $\angle B$ are Complementary Angles, then their measures add to 90°
- If $\angle A$ and $\angle B$ are Supplementary Angles, then their measures add to 180°



the box designates a right (90°) angle

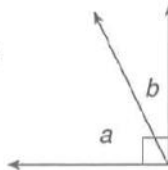
Lines that intersect cross at exactly 1 point. The intersection of two lines creates 4 angles.



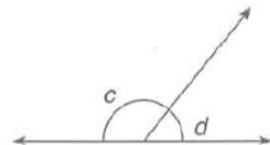
Vertical Angles are opposite each other $\angle 4$ & $\angle 2$ or $\angle 1$ & $\angle 3$

Solve as directed.

1. If $\angle b = 26^\circ$, what is the measure of $\angle a$?

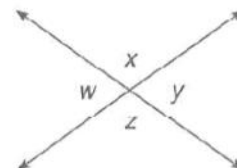


2. If $\angle c = 129^\circ$, what is the measure of $\angle d$?



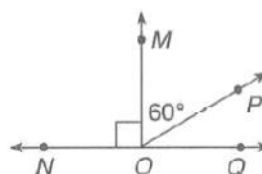
Use the drawing to answer questions 3–5.

3. Name two pairs of vertical angles.
4. If $\angle z$ measures 110° , what does $\angle w$ measure?
5. $\angle z = 110^\circ$. What does $\angle x$ measure? How do you know?



Use the drawing to answer questions 6 and 7.

6. Name a supplementary angle to $\angle MON$.
7. What is the measure of $\angle POQ$?



Polygons

Friday, May 17, 2024 11:49 AM

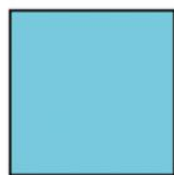
Definition of a Polygon: a plane figure with at least three straight sides, i.e., figures that are made completely of straight lines and the entire figure is closed.

Polygons

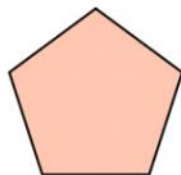
Note: it is redundant to say "straight line" because, by definition, all lines are straight.



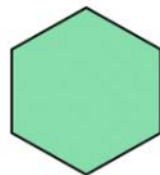
Triangle



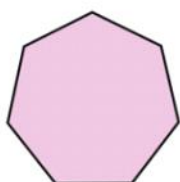
Quadrilateral



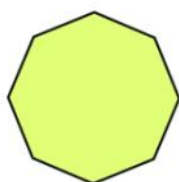
Pentagon



Hexagon



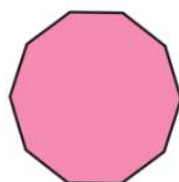
Heptagon



Octagon



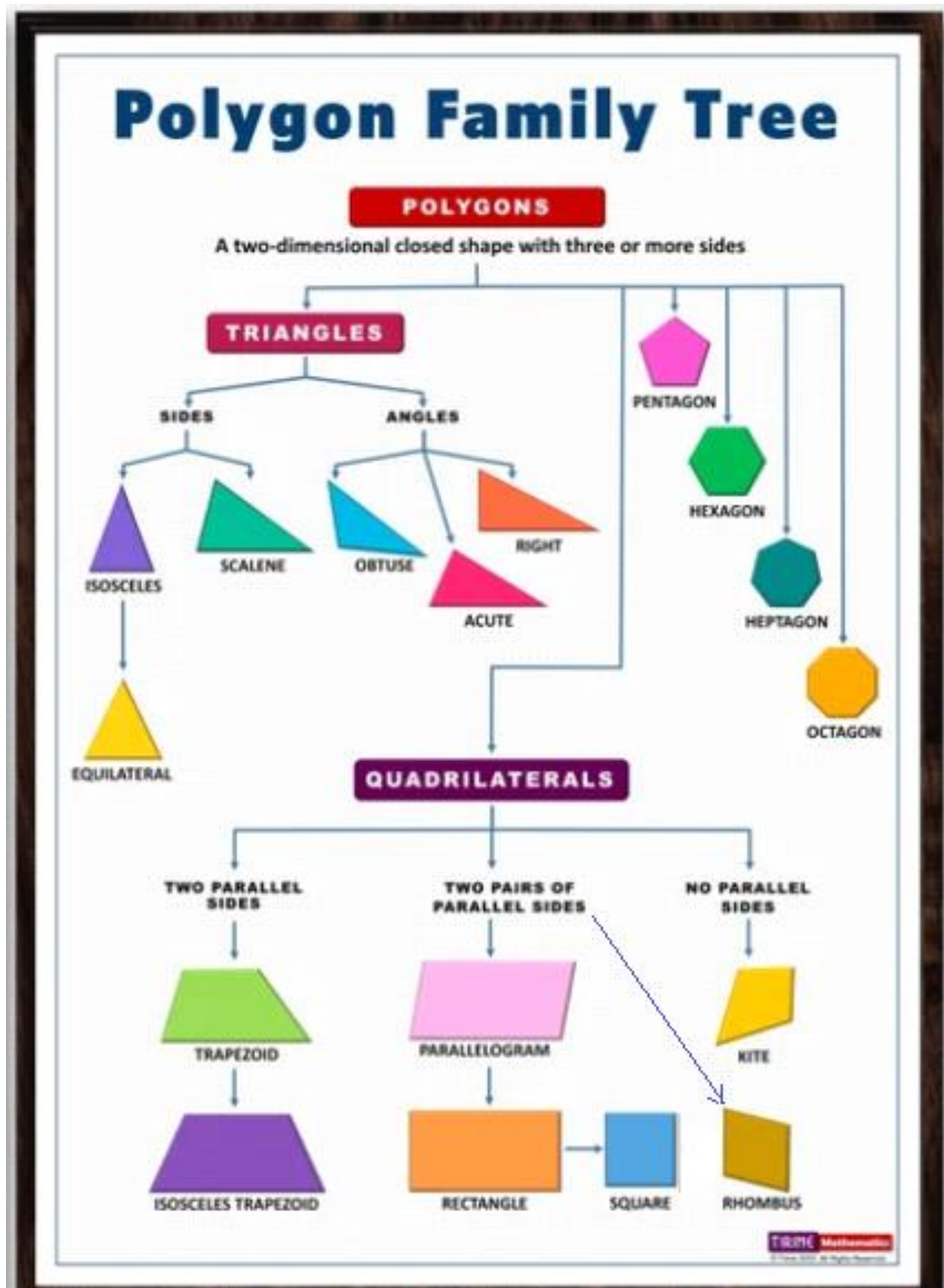
Nonagon



Decagon

Polygon Family Tree

Tuesday, August 27, 2024 9:49 AM



Quadrilaterals

Wednesday, May 29, 2024 7:58 PM

Definition of a Quadrilateral: a polygon with exactly 4 sides. The sum of the inside angles of a quadrilateral is 360° . The points where each pair of lines meet is called a vertex. Quadrilaterals all have exactly 4 vertices.

Definition of a Parallelogram: a parallelogram is a quadrilateral whose opposite sides are parallel and of equal length, and opposite angles are equal in measure.

Definition of a Rectangle: a rectangle is a parallelogram whose opposite sides are parallel and whose inside angles are all 90° each.

Definition of a Square: a square is a rectangle whose sides are all equal.



Parallelogram



Rectangle



Rhombus



Square



Trapezium (UK)
Trapezoid (US)



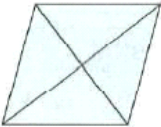
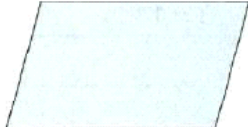
Kite

Quadrilaterals, cont'd...

Wednesday, May 29, 2024

8:06 PM

Below are examples and properties of five common quadrilaterals.

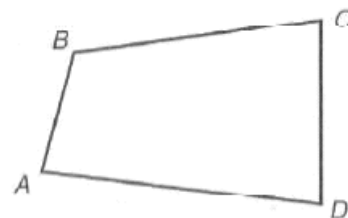
Name	Properties	Examples
square	Opposite sides are parallel. All sides are equal. All angles are right angles (90°). Diagonals are equal and perpendicular.	
rectangle	Opposite sides are equal and parallel. All angles are right angles (90°). Diagonals are equal but not necessarily perpendicular.	
rhombus	Opposite sides are equal and parallel. All sides are equal. Diagonals are perpendicular but not necessarily equal.	
parallelogram	Opposite sides are equal and parallel.	
trapezoid	Only <i>one</i> pair of sides is parallel.	

Quadrilaterals, Examples

Wednesday, May 29, 2024 8:08 PM

Use figure $ABCD$ to answer the questions below.

1. Which side is opposite AB ?
2. Name two sides that are adjacent to BC .
3. Name the two diagonals.
4. Which angle is opposite $\angle ABC$?
5. The measure of $\angle DAB$ is 80° . The measure of $\angle ABC$ is 114° . The remaining two angles are equal in measure. What is the measure of $\angle ADC$?



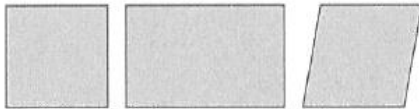
Quadrilaterals, Examples cont'd...

Wednesday, May 29, 2024 8:09 PM

Use the properties of quadrilaterals to answer each question.

6. Name three properties that rectangles and squares have in common.

7. **True or False** A square, a rectangle, and a rhombus are three examples of parallelograms. Explain your answer.

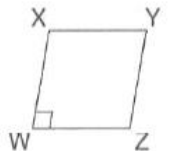


8. A four-sided figure has equal and parallel opposite sides. None of the four angles are right angles, but all four sides are equal in length. What is the name of the figure?

9. **True or False** A trapezoid can *never* have a right angle. Explain your answer.



Imagine you are viewing the quadrilateral to the right from an angle that distorts the figure. You know that WX and YZ are equal, XY and WZ are equal, XW and WZ are equal, and $\angle XWZ$ is a right angle. Use the information to answer the questions below.



10. Is it possible for this figure to have only one right angle?

Why or why not?

12. **Explore** Tiles are often made in the shape of squares because it is easy to fit squares together to cover a surface. Would it be possible to use rhombus-shaped tiles to cover a surface? How about trapezoid tiles?

11. What is the name for this figure?

Triangles

Friday, May 17, 2024 11:49 AM

Definition of a Triangle: a triangle is a polygon with three sides, and the sum of its inside angles is 180° .

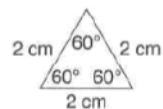
Name

equilateral

Properties

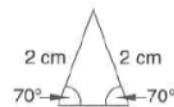
three equal sides
three equal angles

Examples



isosceles

two equal sides
two equal angles



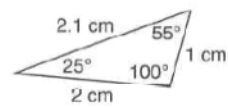
right

one right angle



scalene

no equal sides
no equal angles

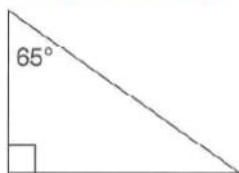


Triangles - Examples

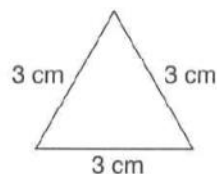
Thursday, May 30, 2024 10:56 AM

Identify each triangle.

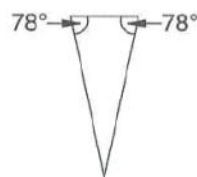
1.



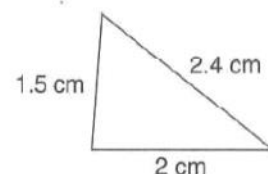
2.



3.



4.

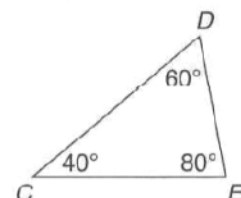


Use your knowledge of triangles to solve problems 5–7.

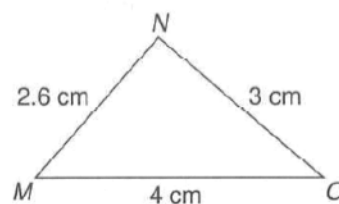
5. Determine which groups of three angles could form triangles.

- _____ $30^\circ, 30^\circ, 90^\circ$
- _____ $50^\circ, 80^\circ, 50^\circ$
- _____ $45^\circ, 60^\circ, 75^\circ$
- _____ $120^\circ, 30^\circ, 45^\circ$
- _____ $50^\circ, 60^\circ, 70^\circ$

6. What is the longest side in triangle CDE ? Why?

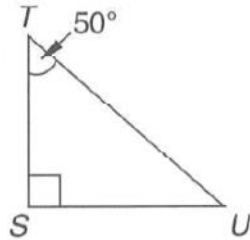


7. What is the largest angle in triangle MNO ? How do you know?

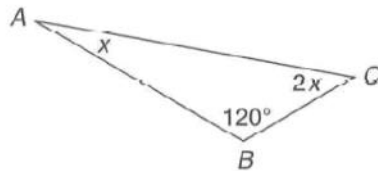


Use algebra to solve for the unknown angle(s).

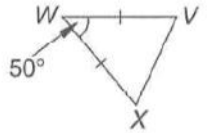
8. What is the measure of $\angle U$?



9. $\angle C$ is twice the measure of $\angle A$. If $\angle B$ measures 120° , what are the measures of $\angle A$ and $\angle C$?



10. In the isosceles triangle VWX , $\angle W$ measures 50° . What are the measures of $\angle V$ and $\angle X$?

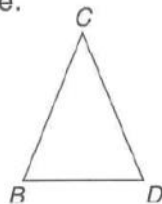


11. **Investigate** The rectangle below is divided into two triangles. Based on your knowledge of triangles, what is the sum of the angles in a rectangle?



Answer these questions about isosceles triangles.

12. Triangle BCD is an isosceles triangle. If sides BC and CD are equal in length, then $\angle B$ must be equal to which other angle?



13. **Draw** Can there be a right angle in an isosceles triangle? Create a drawing that proves your answer.